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**Federated Deep Learning for Prostate Cancer
Gleason
Grading via Histopathology Images and Telehealth
Integration**

Final Year Project Proposal

Submitted By

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1. Introduction

a. Background of the Study

Prostate cancer is one of the most prevalent malignancies among men globally, ranking as the second most common cancer in men worldwide. Early and accurate grading of prostate cancer is critical because it directly governs treatment decisions — from active surveillance of low-risk patients to aggressive intervention for high-grade disease. The Gleason grading system, established by pathologist Donald Gleason, remains the gold standard for histopathological assessment of prostate cancer severity. A Gleason score is assigned by examining the two most dominant architectural growth patterns of prostate gland tissue in a biopsy, scoring each from 1 to 5 based on cellular differentiation, and summing the two scores (ranging from 2 to 10). Higher Gleason scores correspond to more aggressive, poorly differentiated tumors with worse prognoses.

The advent of digital pathology has transformed this traditionally manual and observer-dependent process. Whole-slide images (WSIs) of prostate biopsies — digitized at high resolution — can now be analyzed by deep learning (DL) algorithms capable of matching or exceeding pathologist-level performance on specific grading tasks. Convolutional Neural Networks (CNNs) such as ResNet, EfficientNet, and architectures adapted for gigapixel images have demonstrated strong classification accuracy on benchmark prostate cancer datasets. However, two critical challenges impede deployment of these models in real clinical settings. First, WSIs are extremely high-resolution (often exceeding $100,000 \times 100,000$ pixels), making direct end-to-end learning computationally prohibitive. A sequential, patch-based pipeline — where tissue regions are first localized and then graded at the patch level — is necessary to manage this scale. Second, assembling sufficiently large, diverse training datasets across hospitals raises severe patient privacy and data governance concerns; raw histopathology images cannot be freely shared across institutional boundaries due to regulations such as HIPAA and GDPR.

Federated Learning (FL) offers an elegant solution to the data-sharing dilemma. Instead of centralizing patient data, FL enables multiple hospitals to collaboratively train a shared model by exchanging only encrypted model parameters — never raw images. This distributed paradigm preserves privacy while allowing the global model to benefit from the statistical diversity of multi-site data. Integrating the resulting AI predictions into a telehealth dashboard further closes the clinical loop, enabling remote urologists and pathologists to review AI-generated Gleason grades, annotated heatmaps, and patient risk summaries in a structured, secure environment.

b. Current Issues

- Centralized AI training requires pooling histopathology images across hospitals, violating patient privacy laws (HIPAA, GDPR) and institutional data governance policies.
- Existing prostate cancer AI models are predominantly trained on single-site datasets, leading to poor generalizability across different staining protocols, scanners, and patient demographics.
- Whole-slide images cannot be fed directly into standard CNN architectures; naive single-pass approaches fail due to GPU memory constraints and loss of spatial context at reduced

resolutions.

- Most deployed clinical AI tools lack explainability — pathologists cannot inspect why the model assigned a particular Gleason grade, undermining clinical trust and regulatory compliance.
- There is limited integration between AI-based grading tools and telehealth workflows, creating a gap between algorithmic outputs and actionable clinical decisions.

c. Problem Statement

How can multiple urology centers and hospitals collaboratively train a robust, privacy-preserving deep learning model for prostate cancer Gleason grading from histopathology whole-slide images — using a computationally feasible two-stage patch-based pipeline — while providing explainable grading outputs and integrating results into a clinician-facing telehealth dashboard for remote review and decision support?

d. Proposed Solution

We propose a Federated Deep Learning system built around a Two-Stage Pipeline for histopathology analysis, coordinated by a Federated Learning Aggregation Server, and surfaced through an interactive Telehealth Dashboard. Concretely, the system consists of:

- **Stage 1 — Tissue Localization & Segmentation:** A U-Net or OpenCV-based patch extraction module that uses expert annotation maps (from the Gleason 2019 dataset) to localize cancerous gland regions of interest (ROIs) within WSIs and extract fixed-size patches for downstream classification.
- **Stage 2 — Gleason Grade Classification:** A lightweight CNN (ResNet34 or EfficientNet-B3) that assigns Gleason grades to extracted patches, producing patch-level probability scores aggregated into a slide-level Gleason score prediction.
- **Stage 3 — Federated Learning Simulation:** The Kaggle training dataset is partitioned into 3–4 simulated hospital clients. A custom FedAvg training loop exchanges only model weights — never raw images — across client nodes, producing a globally robust model without data centralization.
- **Module 4 — Telehealth Dashboard:** A web-based dashboard where registered urologists and pathologists can review AI-generated Gleason grades, SHAP/GradCAM explainability heatmaps overlaid on WSI patches, and provide prescription or referral recommendations with mandatory clinician confirmation before any clinical action.

All AI outputs are explicitly labeled as decision support only. No automated prescribing or diagnosis is performed without clinician review and authorization.

2. Existing System / Related Works

The following table summarizes key existing works across the four core pillars of this project: federated learning in medical imaging, Gleason grading with deep learning, histopathology patch-based pipelines, and telehealth decision support systems.

Table 1: Summary of Existing Systems / Work / Research Articles

Article / System	Year	Features	Strengths	Limitations
Federated Learning in Medical Imaging (Rieke et al.)	2020	FL across hospital nodes for privacy-preserving model training	Demonstrates multi-site feasibility and strong privacy guarantees	Focused on segmentation; not applied to Gleason grading or WSIs
PANDA Challenge — Prostate Cancer Grade Assessment (Bulten et al.)	2022	Global Kaggle challenge for AI-based Gleason grading; ensemble CNN models	Strong benchmark; diverse multi-country dataset	Centralized training requires data aggregation; no FL component
Gleason 2019 Dataset & Annotation (Nir et al.)	2019	Expert-annotated WSIs with gland-level Gleason maps; multi-pathologist consensus	High-quality ground truth; enables segmentation + classification pipeline	Single-institution dataset; limited scanner diversity
U-Net for Gland Segmentation (Sirinukunwattana et al.)	2017	Encoder-decoder CNN for tissue region detection in colorectal histopathology	Effective patch localization; generalizable architecture	Not applied to prostate cancer; no federated context
ResNet/EfficientNet for Histopathology Classification	2018–2022	Transfer learning-based patch classifiers for cancer subtype detection	High accuracy; lightweight enough for local training	Single-site training; no privacy-preserving mechanism
FedAvg (McMahan et al.)	2017	Federated Averaging — canonical FL algorithm for distributed model aggregation	Communication-efficient; privacy-preserving by design	Assumes IID data; performance degrades on heterogeneous distributions
Telehealth Decision Support Systems (review)	2019–2023	Clinician dashboards integrating AI outputs for remote triage and consultation	Improves access to specialist review; streamlines workflow	Regulatory hurdles; limited AI explainability integration
SHAP / GradCAM for Explainable Pathology AI	2017–2021	Feature attribution and attention-based heatmaps for CNN predictions	Increases clinician trust; supports regulatory transparency	Computationally expensive at WSI scale; not always clinically intuitive

The reviewed literature demonstrates that while deep learning for Gleason grading has reached near-pathologist accuracy in centralized settings, and federated learning has proven feasible for medical imaging tasks, the intersection of these two paradigms — federated patch-based Gleason

grading with telehealth integration and explainability — remains largely unaddressed. This project directly targets that gap.

3. Proposed System

3.1 Overview

The proposed system is a multi-site Federated Deep Learning framework tailored for prostate cancer Gleason grading from histopathology whole-slide images. Each participating clinic or hospital hosts: (1) local high-resolution WSI data with expert Gleason annotation maps, (2) a two-stage local training pipeline for patch extraction and grade classification, and (3) a secure parameter-exchange client that shares only encrypted model weights with a central aggregation server — never raw patient images.

Registered urologists and pathologists access a secure web-based telehealth dashboard to review AI-generated Gleason grades, explainability heatmaps, and patient case summaries. All AI outputs are framed as clinical decision support; mandatory clinician review and confirmation is enforced before any prescription or referral action is recorded.

3.2 Training Strategy: Two-Stage Pipeline

The Gleason 2019 dataset contains massive, high-resolution slide images and detailed expert annotation maps (Maps1_T through Maps6_T) indicating gland regions and their cancer grades. Feeding these WSIs directly into a standard CNN is computationally infeasible. The system therefore employs a sequential Two-Stage Pipeline:

Stage 1: Tissue Localization and Patch Extraction

The objective of Stage 1 is to identify biologically meaningful regions within each WSI rather than processing the entire image uniformly. The system uses the provided expert annotation maps to train a U-Net segmentation model — or, as a computationally lighter alternative, leverages OpenCV-based thresholding and contour detection — to delineate cancerous gland tissue from background stroma and benign regions. Fixed-size patches (e.g., 256×256 or 512×512 pixels) are extracted from the identified regions of interest (ROIs), discarding uninformative background tiles. Data augmentation operations (random flips, rotations, color jitter for stain normalization) are applied to increase patch diversity and model robustness.

Stage 2: Gleason Grade Classification

The information-rich tissue patches produced by Stage 1 are passed to a lightweight CNN classifier — either ResNet34 or EfficientNet-B3 — pre-trained on ImageNet and fine-tuned for the Gleason grading task. The model outputs per-patch probability distributions across Gleason grade categories (Grades 3, 4, and 5, corresponding to increasing malignancy). These patch-level predictions are aggregated — via majority voting or a learned aggregation head — into a slide-level Gleason score and ISUP grade group, which forms the primary diagnostic output of the system.

Stage 3: Federated Learning Simulation

To simulate a real-world multi-hospital deployment, the Kaggle training image set is divided into 3–4 non-overlapping partitions, each assigned to a simulated hospital client (implemented as distinct PyTorch DataLoaders). A custom federated training loop implements the Federated Averaging

(FedAvg) algorithm:

- Each client independently trains the two-stage model on its local data partition for a fixed number of local epochs.
- After local training, only the serialized model parameters (weights) — not raw image patches — are transmitted to the central aggregation server.
- The server computes a weighted average of received parameters (weighted by number of local samples) and broadcasts the updated global model back to all clients.
- This communication round is repeated for multiple global rounds until convergence.

No raw patient data leaves any simulated hospital node at any point. The framework is implemented using Python-based FL libraries (PySyft, TensorFlow Federated, or FedML) with gRPC/HTTPS secure channels for parameter exchange.

3.3 System Architecture and Modules

Module A: Data Ingestion and Local Preprocessing

- Load WSI TIFF files and corresponding annotation maps (Maps1_T through Maps6_T) from the Gleason 2019 / Kaggle PANDA dataset.
- Apply stain normalization (Macenko or Reinhard method) to reduce inter-scanner color variation.
- Generate patch libraries per client node; store with associated Gleason grade labels derived from annotation maps.
- Maintain a local metadata registry (slide ID, scanner type, patient age group) without sharing patient identifiers.

Module B: Stage 1 — Segmentation and ROI Localization

- U-Net or OpenCV-based pipeline to segment glandular tissue and exclude background, adipose, and benign stroma.
- Output: binary or multi-class segmentation masks; bounding boxes for ROI patch extraction.
- Patch extraction at multiple magnification levels (e.g., 10× and 20×) to capture both architectural and cytological features.

Module C: Stage 2 — CNN Gleason Grade Classifier

- Backbone: ResNet34 or EfficientNet-B3, initialized with ImageNet weights.
- Fine-tuned for 5-class or 3-class Gleason grade classification (Grades 3/4/5 or ISUP groups 1–5).
- Output per patch: grade probability vector. Slide-level aggregation via majority vote or attention-based pooling.
- Loss function: weighted cross-entropy to handle class imbalance (Gleason 5 is rarer than Gleason 3).

Module D: Federated Learning Coordinator

- Central server manages global model state and orchestrates communication rounds.
- Each client node runs local training (Modules A–C) and submits parameter updates.
- Server applies FedAvg (or optionally FedProx for heterogeneous data distributions) to merge updates.
- Encrypted parameter transmission via gRPC/HTTPS with TLS; no raw data ever leaves client nodes.
- Differential privacy noise injection (optional) for additional privacy guarantees per McMahan et al.

Module E: Explainability and Clinical Safety

- GradCAM and SHAP applied to CNN classifier to generate heatmaps highlighting high-grade tissue regions within patches.
- Heatmaps are overlaid onto the original WSI viewer for pathologist review.
- Confidence scores and uncertainty estimates (Monte Carlo Dropout) accompany every prediction.
- Mandatory clinician confirmation step enforced in the dashboard before any clinical recommendation is recorded.
- Visible disclaimer on all AI outputs: 'This result is AI-generated decision support only. Clinical diagnosis requires pathologist authorization.'

Module F: Telehealth Dashboard and Urologist/Pathologist Registry

- Secure web dashboard built with React (full deployment) or Streamlit (rapid prototype).
- Case list with AI-assigned Gleason scores, confidence levels, and flagged high-risk cases.
- Integrated WSI viewer with GradCAM heatmap overlay for spatial explainability.
- Role-based access control (RBAC): patient data visible only to assigned clinicians.
- Consultation scheduling, prescription recording, and referral recommendation tools with full audit trail.
- Authentication via OAuth2/JWT; encrypted sessions over TLS.

3.4 Scope

The project targets a prototype system demonstrating: (1) a functional two-stage Gleason grading pipeline trained and evaluated on the Gleason 2019 / Kaggle PANDA dataset; (2) a federated training simulation across $N = 3\text{--}4$ data partitions representing distinct hospital clients; (3) explainability outputs (GradCAM heatmaps, SHAP feature attributions) for all model predictions; and (4) an interactive clinician dashboard with role-based access, case review functionality, and audit logging. Real clinical partner deployment is outside the current scope but is identified as the primary future work direction.

3.5 Advantages

- Preserves patient privacy: raw histopathology images never leave local hospital nodes.

- Two-stage pipeline makes WSI analysis computationally tractable on consumer-grade GPUs.
- Federated training leverages multi-site data diversity without data centralization.
- Explainability heatmaps and confidence scores increase clinician trust and regulatory compliance.
- Telehealth dashboard integrates AI grading into clinical workflows, enabling remote pathologist review.
- Modular architecture is extensible to additional cancer types, imaging modalities, and hospital sites.

3.6 Limitations

- Federated simulation on a single machine does not fully replicate real-world network latency and node heterogeneity.
- Performance of FedAvg degrades with highly non-IID data distributions across hospital nodes.
- Regulatory and ethics approvals (institutional review board clearance) required for real patient data deployment.
- Stain normalization may not fully eliminate inter-scanner variability in diverse multi-site deployments.
- Computational resources (high-VRAM GPU) required for U-Net training and GradCAM computation at full WSI scale.

4. Hardware / Software Requirements

Hardware Requirements

- **GPU:** NVIDIA RTX 3090 / RTX 4080 or equivalent (minimum 16 GB VRAM) for U-Net segmentation training and CNN fine-tuning; cloud GPU (Kaggle P100/T4, Google Colab A100, or AWS p3 instance) acceptable for prototype.
- **CPU:** Intel Core i7 / AMD Ryzen 7 (8+ cores) for parallel patch extraction and preprocessing.
- **RAM:** Minimum 32 GB system RAM recommended for loading WSI metadata and patch libraries; 16 GB minimum viable.
- **Storage:** Minimum 200 GB SSD for raw WSI tiles, patch libraries, model checkpoints, and experiment logs.
- **Network:** Reliable high-speed internet connection for federated parameter exchange and dashboard access.

Software Requirements

Category	Tools / Libraries
Core Language	Python 3.10+
Deep Learning Frameworks	PyTorch 2.x, TensorFlow 2.x
Federated Learning Libraries	PySyft (OpenMined), TensorFlow Federated, FedML, OpenFL (Intel)
CNN Architectures / Pretrained Models	torchvision (ResNet34, EfficientNet-B3), timm library
Segmentation (Stage 1)	segmentation-models-pytorch (U-Net), OpenCV 4.x
Histopathology / WSI Tools	OpenSlide (WSI reading), TIAToolbox, QuPath (annotation review)
Explainability	SHAP, GradCAM (pytorch-grad-cam), Captum (PyTorch)
Data Processing	NumPy, Pandas, scikit-learn, scikit-image, Albumentations (augmentation)
Backend API	FastAPI or Flask (REST endpoints for dashboard)
Frontend Dashboard	React.js (full deployment) and/or Streamlit / Gradio (rapid prototype)
Communication Security	gRPC, HTTPS / TLS, OAuth2 / JWT for authentication
Database	PostgreSQL or MongoDB (case logs, clinician records, audit trails)
Experiment Tracking	Weights & Biases (W&B;) or MLflow for training metrics and model versioning
Dataset	Gleason 2019 Dataset (Nir et al.) and/or Kaggle PANDA Challenge Dataset

Clinical/Legal Note: *The system will explicitly label all AI outputs as clinical decision support only. Automated Gleason grading will not substitute for pathologist review. No prescription or clinical recommendation is recorded without mandatory clinician authorization. The system will display a persistent disclaimer on all AI result screens.*

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